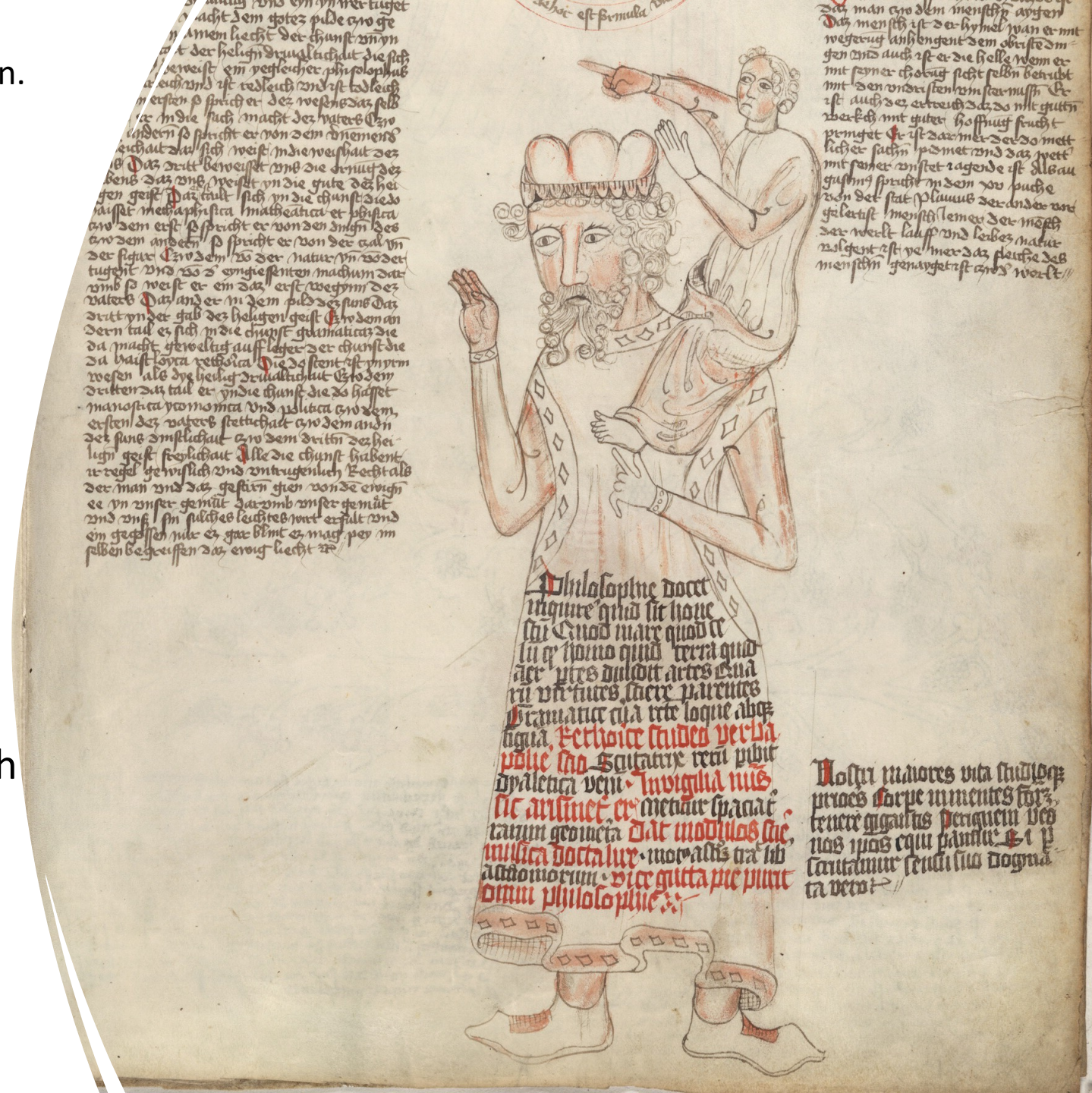


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Science of Science

The Science Behind Scientific Research
Social Studies of Science
Tutorial #3
Xuanlong QIN



Today

- How are scientific breakthroughs made?
- What factors contribute to a scientist's productivity and impact?
- How do collaborations and networks influence scientific outcomes?

Standing
on the
shoulders
of giants

Google Scholar



☒ Articles ☐ Case law

Stand on the shoulders of giants

The phrase "**standing on the shoulders of giants**" is a [metaphor](#) which means "using the understanding gained by major thinkers who have gone before in order to make intellectual progress"

Knowledge Big Bang

- FUN fact with scientific research
 - Publication Growth: Scientific output has surged from doubling every 15 years in the 20th century to a rate of over 3 million articles published annually with a growth rate of 3-4%. (International Association of Scientific, Technical and Medical Publishers; 2018)
 - Patent Applications: Global patent applications grew from about 1 million to over 3.3 million from early 1980s to 2018. (World Intellectual Property Organization;2018)

- Growth of Digital Data: A report by IDC predicts that the global datasphere will grow from 33 zettabytes (ZB) in 2018 to 175 ZB by 2025, much of it driven by scientific and technological advancements.
- Expansion of Scientific Disciplines: the Physics and Astronomy Classification Scheme (PACS) used by the American Institute of Physics included over 6,000 categories as of 2010.

Really?

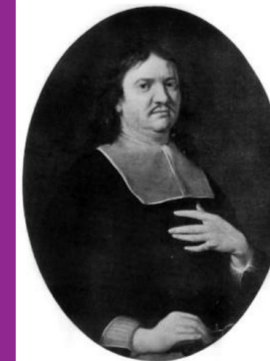
- We have witnessed exponential growth in the volume of new scientific and technological knowledge, thereby creating conditions that should be ripe for major advances.



The STM Report

An overview of scientific and scholarly journal publishing

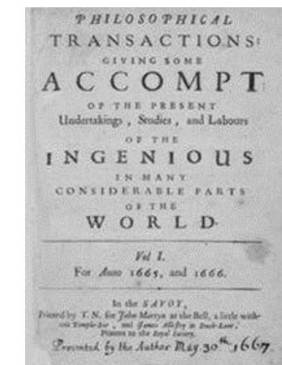
Celebrating the 350th anniversary of journal publishing



Mark Ware
Mark Ware Consulting

Michael Mabe
International Association of Scientific,
Technical and Medical Publishers

Fourth Edition
March 2015



Moore's law is Ending?

- Moore's law is the observation that the number of transistors in an integrated circuit (IC) doubles about every two years. Moore's law is an ***observation*** and ***projection*** of a historical trend. Rather than a law of physics, it is an *empirical relationship linked to gains from experience in production*.
- [\[Video\]](#)
- How about Scientific Innovation?

nature

NET LOSS

Papers and patents are becoming less disruptive of the scientific status quo

Social upheaval
Will the turmoil engulfing Twitter also disrupt science?

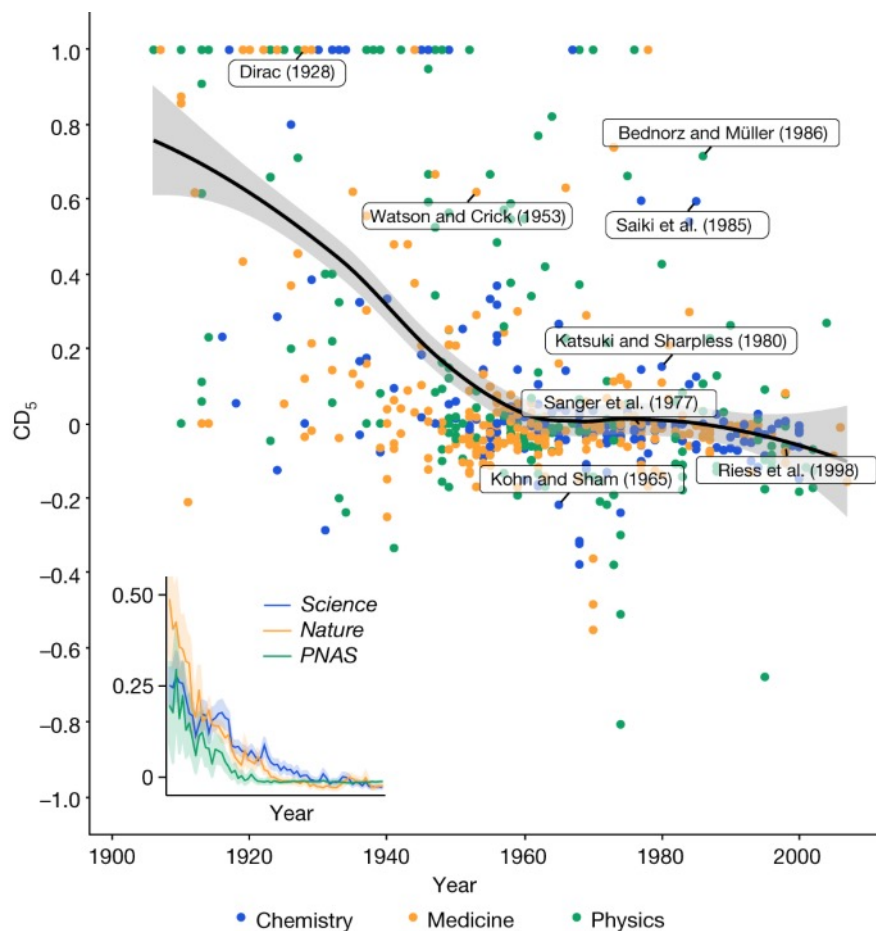
Evolutionary odyssey
The forgotten legacy of Alfred Russel Wallace's voyage to the Amazon

Farming life
Eleven ways to help croplands reduce nitrogen pollution

Vol. 612 No. 7942
nature.com

Papers and patents are becoming less disruptive over time (Nature 2023; Michael Park et al.)

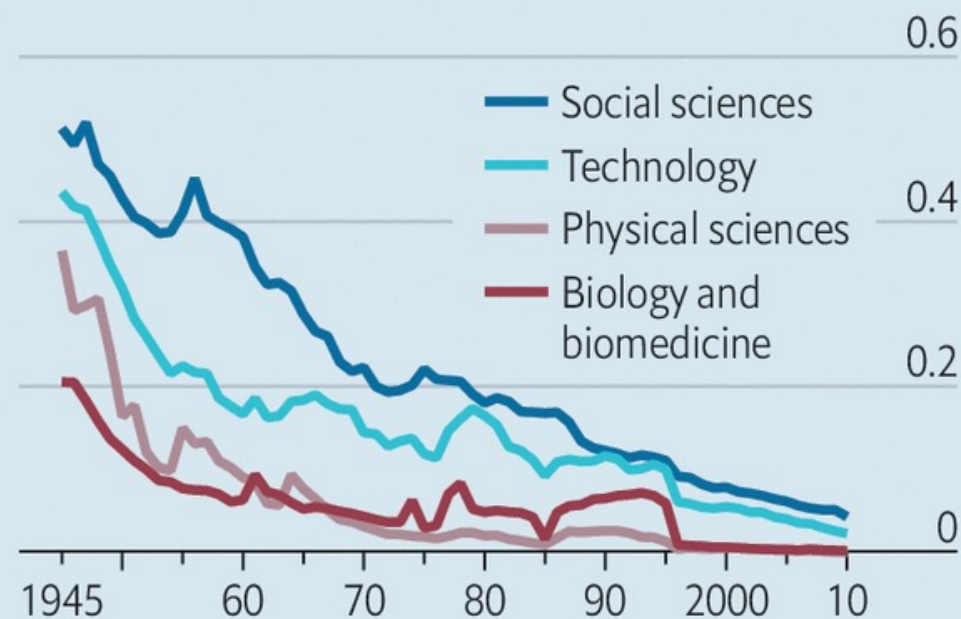
- Why that is, is a mystery
 - “Ideas are like rabbits,” John Steinbeck said. “You get a couple and learn how to handle them, and pretty soon you have a dozen.” Scientific and technological progress is often viewed in this way.



Dirac (1928): Discovery of the Dirac equation of relativistic quantum theory
 Watson and Crick (1953): Discovery of the structure of the DNA
 Kohn and Sham (1965): Development of a new method for calculating electronic structure
 Sanger et al. (1977): Development of a new method for mapping the order of nucleotides
 Katsuki and Sharpless (1980): Development of asymmetric epoxidation
 Saiki et al. (1985): Discovery of polymerase chain reaction
 Bednorz and Müller (1986): Discovery of superconductivity in ceramic materials
 Riess et al. (1998): Discovery of the accelerating expansion of the universe

Decline and fall?

Impact of research papers*, by category
 1=max disruption[†]



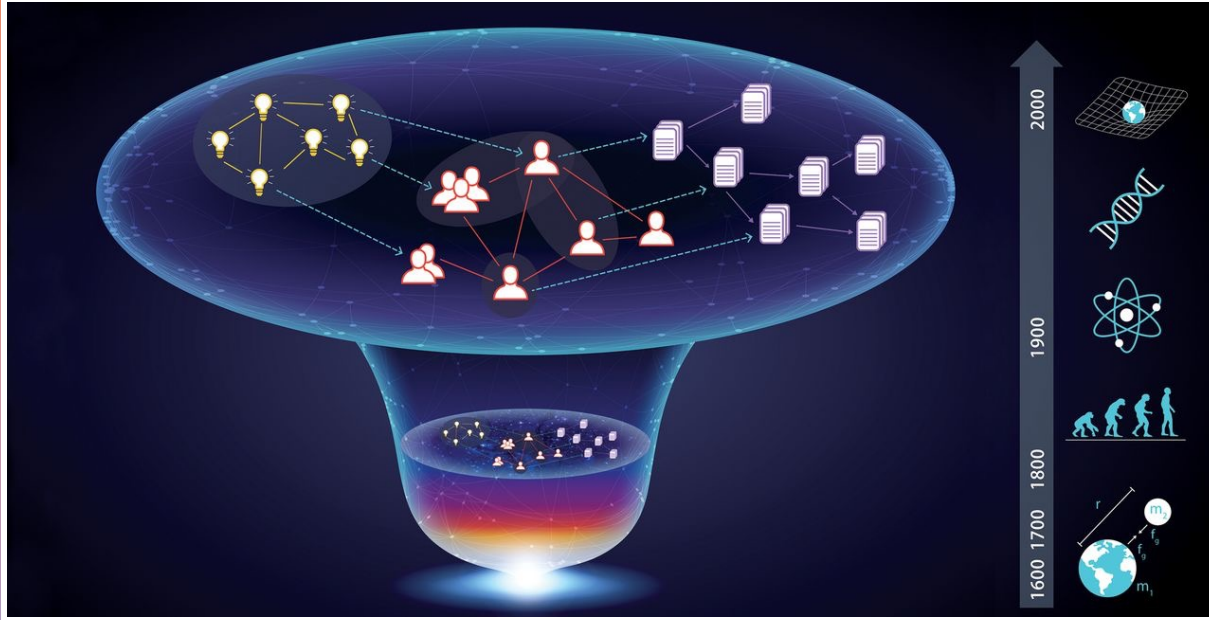
*Five years after publication

[†]All subsequent work does not cite predecessors' work

Source: "Papers and patents are becoming less disruptive over time", by M. Park, E. Leahey & R. J. Funk, *Nature*, Jan 2023

We find...

- Researchers could use scientific methodologies to understand and predict scientific discovery.
- Go further
- How? Is it available?

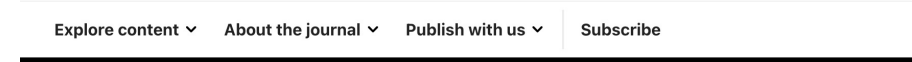


Definition of "Science of Science"

- "A science that uses large-scale data on research production to search for universal and domain-specific patterns."
- The science of science aims to maximize individuals' odds of success.

The complexity of science.

Science can be seen as an expanding and evolving network of ideas, scholars, and papers. SciSci searches for universal and domain-specific laws underlying the structure and dynamics of science



BOOK REVIEW | 03 May 2021

Big data bibliometrics must take into account qualitative analyses of research as a social institution, rooted in history, economics and politics.





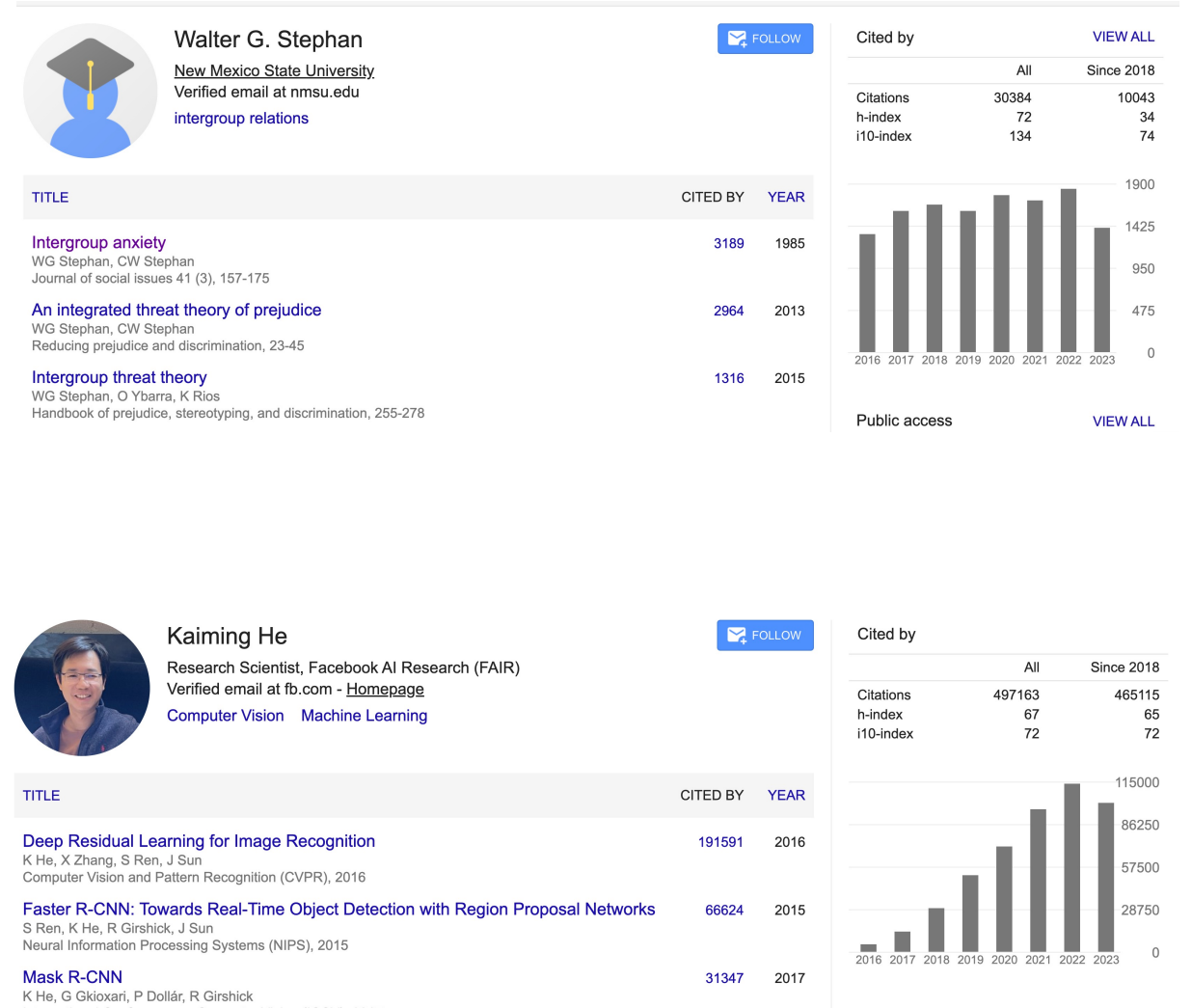


Start with three key concepts

- Productivity
- Impact
- Novelty

Discussion

- Productivity: the number of scientific outputs produced by an individual or a team.
- How could we know a scientist's productivity?
- Measurement?



Case study – H Index

the smartest physicists ever

Landau genius scale	
0	Isaac Newton
0.5	Albert Einstein
1	Paul Dirac Erwin Schrodinger Richard Feynman [11] Niels Bohr Werner Heisenberg Louis de Broglie [11] Satyendra Bose Eugene Wigner , and a few others
1.5	Lev Landau (in later years) [8] Hans Bethe (2016) [12]
2	Lev Landau (after 1962 Nobel Prize) Juan Maldacena (2008) [9]
2.5	Lev Landau (before 1962 Nobel Prize)
3	Edward Witten (2008) [9]
3.5	
4	
4.5	David Mermin (own estimate $\pm$)
5	Mundane or "pathologists"



A measurable indicator

H - Index

The h -index is defined as the maximum value of h such that the given author/journal has published at least h papers that have each been cited at least h times

"Lev Landau's Genius Scale"

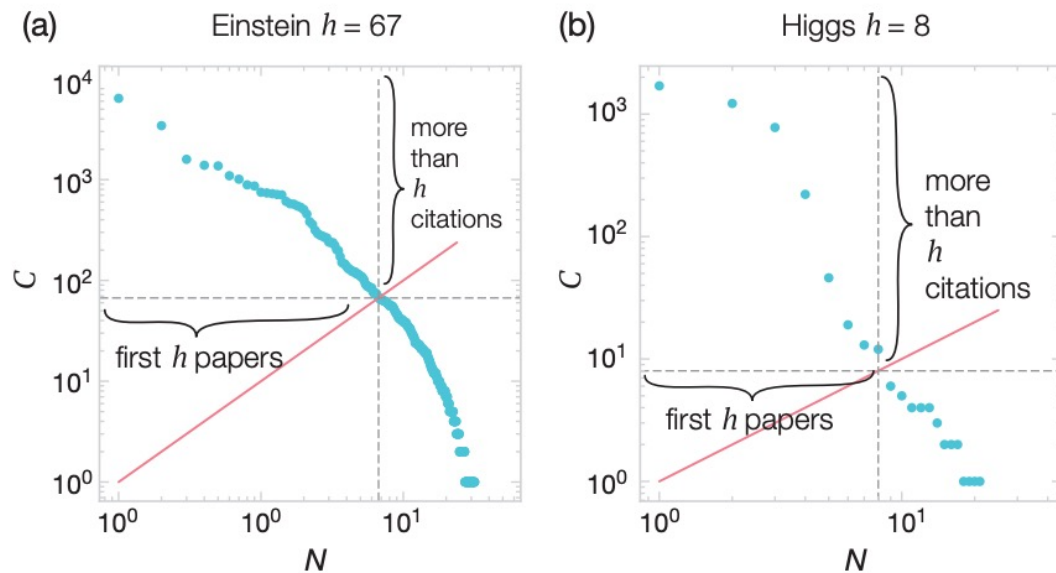


Figure 2.1 The h -index of Albert Einstein (a) and Peter Higgs (b). To calculate the h -index, we plot the number of citations versus paper number, with papers listed in order of decreasing citations. The intersection of the 45° line with the curve gives h . The total number of citations is the area under the curve [26]. According to Microsoft Academic Graph, Einstein has an h -index of 67, and Higgs 8. The top three most cited papers by Einstein are: (1) Can quantum mechanical description of physical reality be considered complete, *Physical Review*, 1935; (2) Investigations on the theory of Brownian movement, *Annalen der Physik*, 1905; and (3) On the electrodynamics of moving bodies, *Annalen der Physik*, 1905. The top three for Higgs are: (1) Broken symmetries and the masses of gauge bosons, *Physical Review Letters*, 1964; (2) Broken symmetries, massless particles and gauge fields, *Physics Letters*, 1964; (3) Spontaneous symmetry breakdown without massless bosons, *Physical Review*, 1966.

Limitation:

1. **Time Insensitivity:** The h -index does not account for how long a researcher has been active. It usually favors those who have been in the field longer, as they've had more time to accumulate citations.
2. **Field Dependency:** Citation practices vary greatly among different fields. Some fields tend to have more citations than others, leading to higher h -indices for researchers in those fields. This makes cross-disciplinary comparisons challenging.
3. **Ignores Citation Distribution:** The h -index does not account for the distribution of citations across a researcher's publications. A researcher could have a few highly-cited papers and many rarely-cited ones, or a more even citation distribution, and still have the same h -index.

Goodhart's law

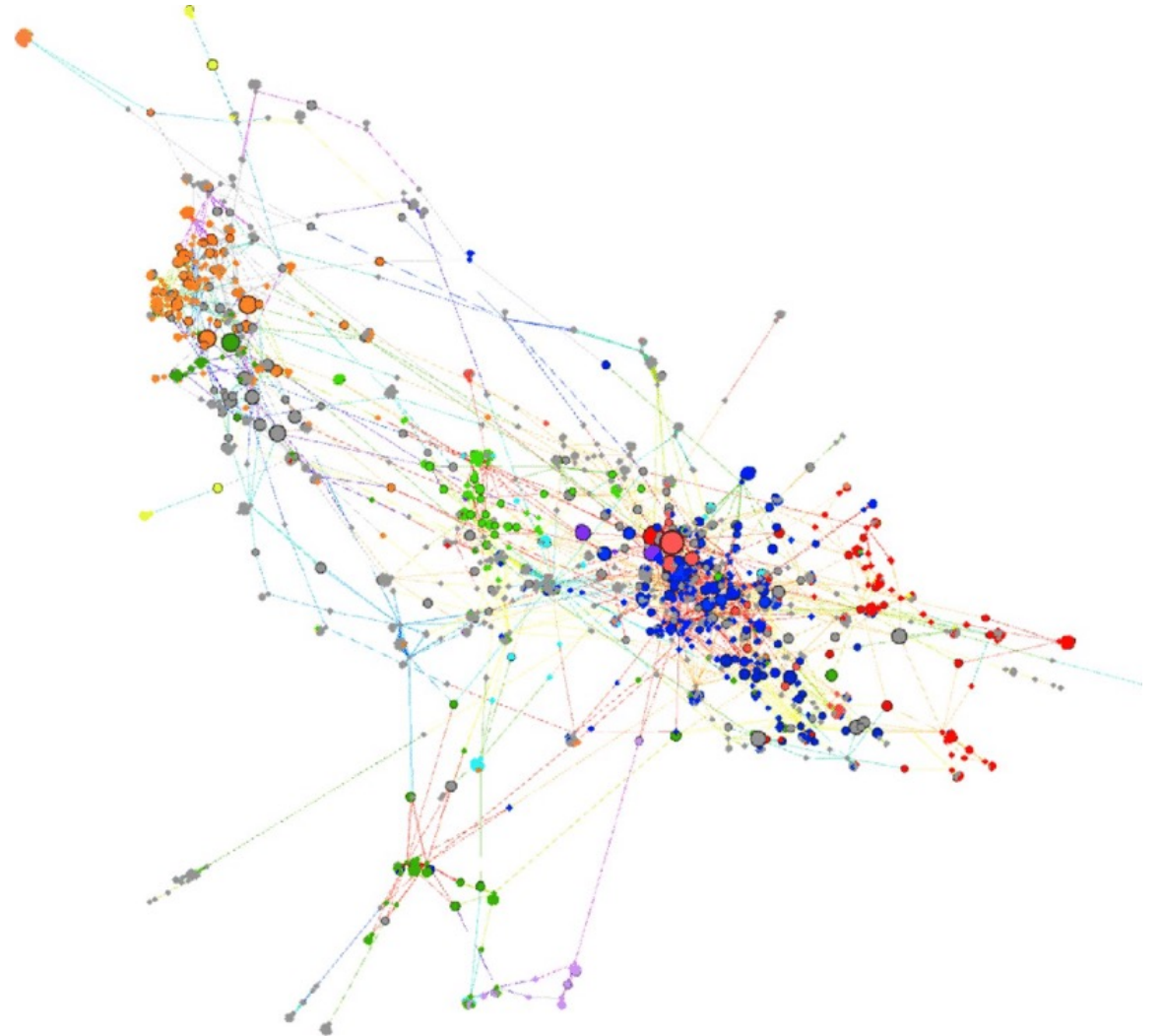
- “When a measure becomes a target, it ceases to be a good measure.”
- Mario Biagioli related the concept to the consequences of using citation impact measures to estimate the importance of scientific publications:
 - “All metrics of scientific evaluation are bound to be abused. When a feature is picked as an indicator of the scientific evaluation, then it inexorably ceases to function as that indicator because people start to game it.”

Debating

- Margit Osterloh and Bruno Frey call it a “taste for rankings”, rather than a “taste for science”.
- These concerns have been promoted through consensus statements such as the << Leiden Manifesto >> and the <<Declaration on Research Assessment>>, which have been signed by thousands of institutions and more than 17,000 individuals.

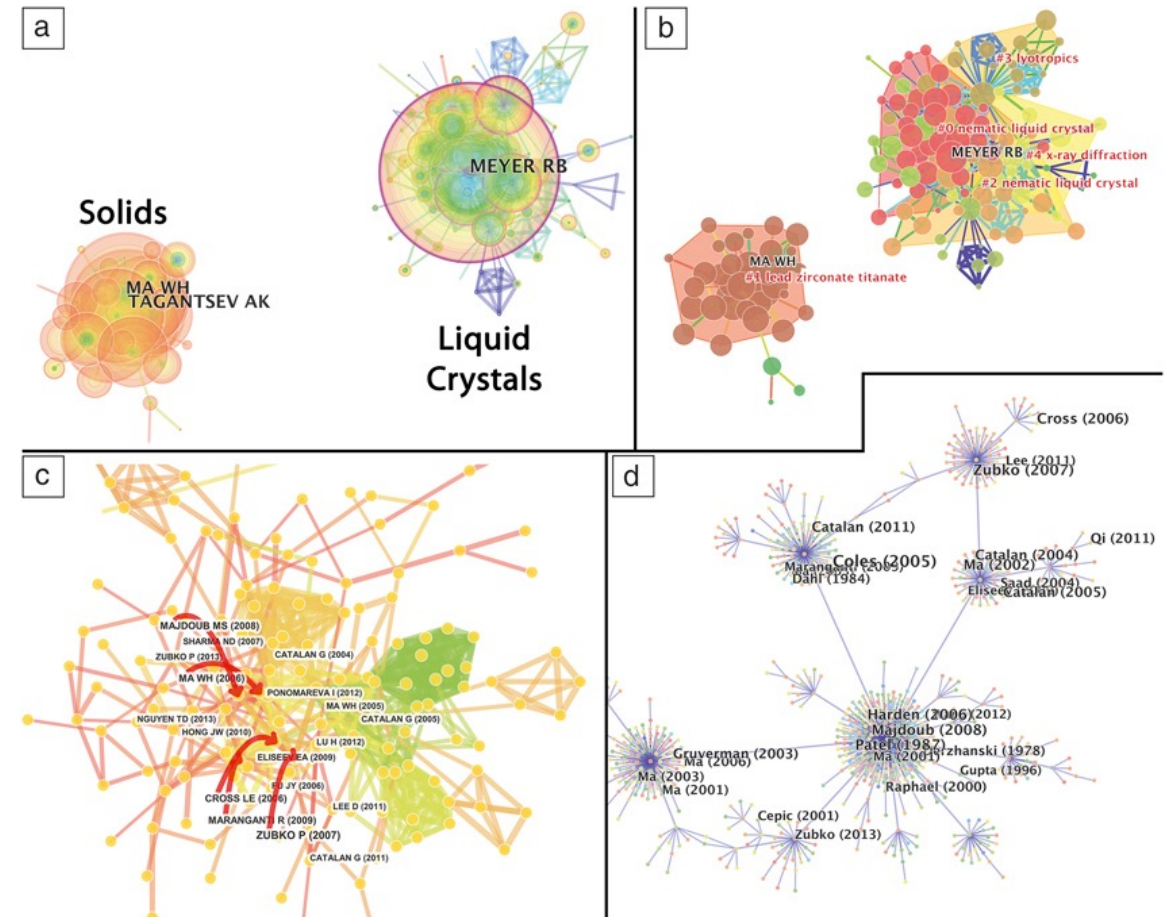
Discussion

- Impact: influence or change brought about by a scientific work
- How impact can be quantified and predicted?
- Measurement?



Impact

- Common methods and metrics used in this process
- Take into consideration
 - Citation Count
 - Journal Impact Factor (JIF)
 - H-Index
 - Altmetrics
 - Author Reputation
 - Timeliness
 - ...



Flexoelectricity CiteSpace analysis. (a) Author networks. clearly showing two sub-networks distinguishing the two distinct components to this field. (b) Clustering analysis, showing that the distinction between the members of the two major author networks is related to the material type (one a ceramic and the other liquid crystals). Each cluster is a different color. (c) Within each network, we can look at the cited authors and the links between them. Shown here is the network titled "solids" in (a). (d) Network of cited papers from within the "solids" citation network.

Case study – Citation network

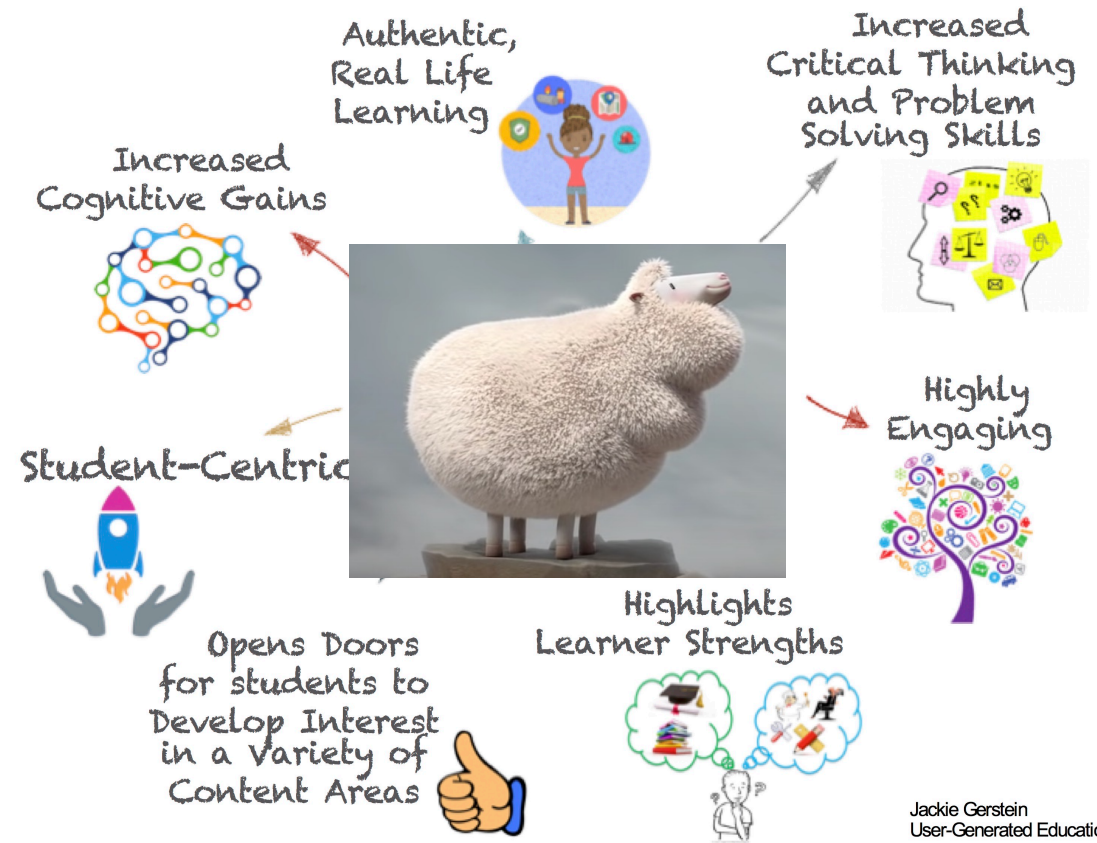
- How Are Ideas Spread at Your Organization?
 - Analyze Its Citation Network!
- Citation is both an impartial scholarly method and a powerful form of social communication.
 - Persuasive citation may be necessary to communicate new, sound claims to the scientific community;
 - Generally speaking, research texts that receive more citations from others symbolize a degree of prominence to a field of study;

Debating

- How **citation distortions** create unfounded authority: analysis of a citation network?
 - Authority and belief in a claim specific citation network
 - **Citation bias** against critical primary data
 - Amplification through influential papers and citations
 - Network authority emerges through bias against critical content and amplification
 - Systematic ignoring of papers that contain content conflicting with a claim

Discussion

- Novelty: the degree of newness or innovation in a scientific work
- [\[video\]](#)
- Discuss the role of novelty in scientific success
- Measurement?



Novelty

- How Novelty is an essential ingredient in scientific discovery, driving the frontiers of knowledge forward
 - Novel Concepts and Theories - Theory
 - Novel Experimental Techniques and Instruments - Methods
 - Novel Applications of Existing Knowledge - Application
 - Novel Interactions and Collaborations – Interdisciplinary
- How does Novelty work appear?

How likely is it
that Einstein's
exceptional
output may
emerge purely
by chance?



Scientific careers are not random

- If it is...
- The most famous scientist will be the person who published the most papers.
- Novelty could be predicated by the Q-model



Theodore Dru Alison Cockerell (1866–1948) was an entomologist and [systematic biologist](#) who published nearly 4,000 papers, some of them only a few lines long.

The Q-model

- Definition of the Q model, as a measure of an individual's ability to make high-impact contributions.
 - it is derived from a scientist's publication record, taking into account both the number of publications (productivity) and their citation counts (impact).

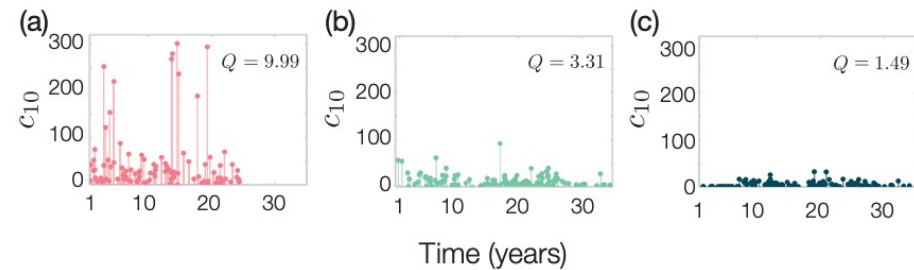


Figure 6.2 Careers with different Q -factor. The figures illustrate career histories of three scientists with comparable productivity ($N \simeq 100$), documenting the notable differences between the papers published by them, given their different Q -factors.

The Q Model and Individual Performance

- Findings(Quantifying Long-Term Scientific Impact, Dashun Wang;2013;Science):
 - Career Longevity and Productivity.
 - They found that productivity, or the number of papers a scientist produces, increases rapidly early in the career, reaches a peak, and then gradually declines.
 - Q Factor
 - An inherent characteristic of a scientist that quantifies their ability to produce high-impact work, a higher Q factor consistently produces higher impact work. Importantly
 - Predictive Power
 - The Q factor could be used as a predictive tool in hiring decisions and funding allocations.

The Q Model and Team Performance

- However, science is often a collaborative endeavor..
- Synergy and Interactions
 - Teams are more than just the sum of their parts. The interaction between team members can lead to synergies that enhance (or sometimes diminish) the team's performance.
- Leadership and Roles
 - Teams often have leaders and members may have specific roles that influence the team's performance. (Lead-ratio)
- Team Size
 - The size of the team might also influence its performance. Larger teams may have more resources and a broader range of skills, but they may also face challenges with coordination and communication
- ...

Case study: Team size

- The growing size of team



Productivity

Collaboration

Interdisciplinary



Is bigger always better?

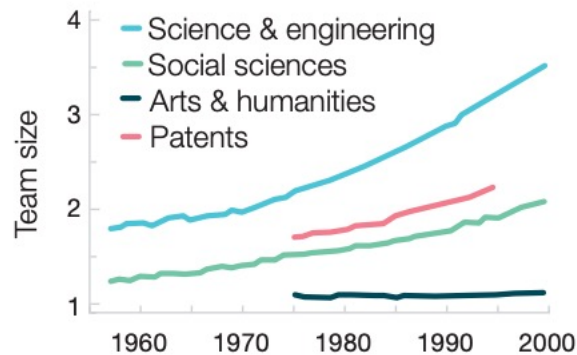


Figure 12.1 The growing size of teams. Each line represents the arithmetic average of team size taken over all subfields in each year. After Wuchty et al. [136].

2 Pizza rule

- According to Bezos, the ideal is the “two pizza team:” if a team couldn’t be fed with two pizzas, it was too big.
- [\[video\]](#)

Case study: Team size

- large teams develop science, whereas small ones disrupt it.

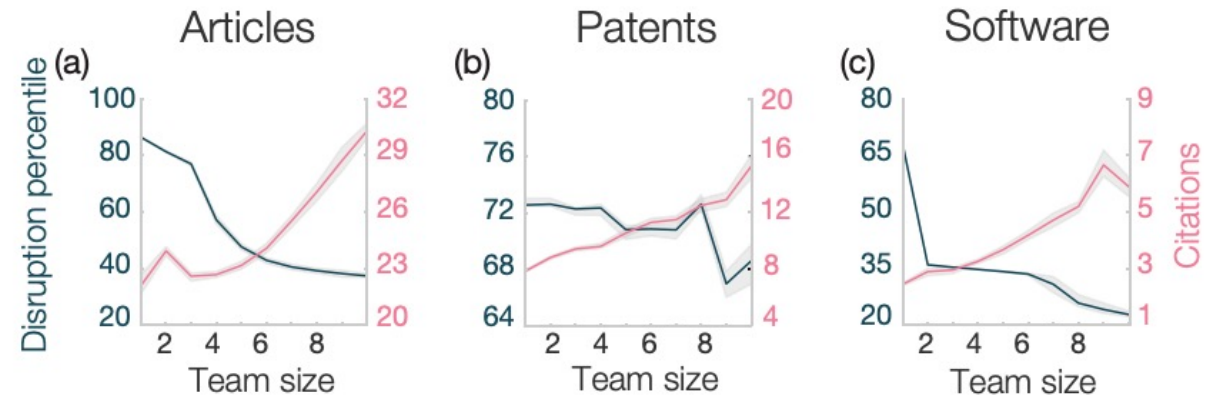


Figure 12.4 **Small teams disrupt, big teams develop.** To examine the relationship between team size and disruption, we collected data from three domains: (1) the Web of Science database, containing articles and the papers they cite; (2) patents granted by the USPTO and their citations; (3) software projects on GitHub, a web platform that allows users to collaborate on software and “cite” other repositories by building on their code. For research articles (a), patents (b), and software (c), median citations (red curves indexed by the right y-axis) increase with team size, whereas average disruption percentile (green curves indexed by the left y-axis) decreases with it. Teams of between 1 and 10 authors account for 98% of articles, 99% of patents, and 99% of code repositories. Bootstrapped 95% confidence intervals are shown as gray zones. After Wu et al. [206].

Debating Q-model

- They then advocate the Q-factor, which seeks to define and predict scientific careers by quantifying an individual's ability to turn an idea into a discovery with a given citation impact. This rests on the assumption that all scientists have access to the same resources, ignoring the massive disparities across countries and institutions.

Practice application

- Actually, there are potential applications of these concepts in real-world settings, such as research management, funding decisions, and career planning.
- Advantage:
 - Efficiency and Scalability
 - Objectivity
 - Accountability and Transparency
 - ...

Criticisms and Controversies

- Bibliometrics(e.g. Scisci) must take into account that science is a social institution.
 - Science is rooted in historical, economic and political environments, and big data bibliometrics is no exception. If the workforce is classified as either superstars or failures, cumulative scholarship is lost.
 - Science does not happen in a vacuum. It is a social and intellectual institution, rooted in historical, economic and political contexts. Underplaying these elements has grave consequences

Criticisms and Controversies

- Imperfect Measures:
 - Metrics such as citation counts or journal impact factors are imperfect measures of research quality or impact
- Over-reliance on Quantification:
 - The use of metrics can lead to an overemphasis on quantifiable outputs, potentially neglecting other important aspects of scientific work, such as teaching, mentorship, service, or impact on policy or practice.

- Hope you can gain something...